

Memo 03: MP1 Mass-Conservation Check and Updated Diagnosis

FVCOM-MP Delaware Test Cases

April 30, 2026

Purpose

This memo updates the MP1 mass-conservation diagnosis after rerunning case **a** with `SEDIMENT_MODEL=T`. The central comparison is now: case **a** as no-floc plastic with sediment runtime on, versus case **b** as floc-enabled but effectively null-kinetic/split-disaggregated plastic.

Archived Analysis Outputs

Three mass-conservation snapshots are now preserved under `FVCOM_MP_test_run/PYTHON/output/mass_conservation/archive/`.

Version	Archive label	Meaning
v001	v001_before_case_a_sediment_on	old comparison with case a using <code>SEDIMENT_MODEL=F</code>
v002	v002_case_a_sediment_on_full10d	comparison with case a using <code>SEDIMENT_MODEL=T</code> and the original sediment BBL settings
v003	v003_case_a_sediment_on_no_bbl	case-a diagnostic with <code>SEDIMENT_MODEL=T</code> but sediment BBL roughness switches disabled

The live analysis folder has also been regenerated from the v003 state. The archived files include `mass_summary.csv`, `mass_timeseries_long.csv`, and all current comparison plots.

Current Evidence

All three current MAT files cover the same FVCOM model-day window, 58485.0–58494.957031, or comparison day 0.00–9.96.

Case	Sediment runtime	Final water (kg)	Final bed (kg)	Final total (kg)	Retained vs. a (%)
a	on, BBL off	573.28	0.00	573.28	100.0
b	on	371.85	0.00	371.85	64.9
c	on	504.26	24.29	528.55	92.2

The major finding from v002 remains that enabling sediment runtime in case **a** reduced the final domain MP1 mass from the old archived value 1001.25 kg to 582.41 kg. The new v003 BBL-off

diagnostic ends at 573.28 kg, only 9.13 kg below the v002 BBL-on value. Therefore a large part of the old case-a/case-b gap was not caused by the split flocc path alone, and the BBL-off test suggests the sediment-runtime effect is not primarily controlled by sediment BBL roughness.

The remaining null-flocc mismatch is still meaningful. With sediment runtime matched in the latest v003 comparison, case b ends 201.43 kg below case a, retaining 64.9% of the sediment-on no-flocc baseline. Case c ends 44.73 kg below case a, retaining 92.2%, with 24.29 kg stored in the bed.

New Negative Test: BBL-Off Case a

To test whether sediment affected plastic only through sediment-controlled bottom roughness, a case-a diagnostic sediment input was prepared: `generic_sediment_a_no_bbl.inp`. This keeps `SEDIMENT_MODEL=T` and suspended sediment transport enabled, but sets `MB_BBL_USE=F`, `SG_BBL_USE=F`, `SSW_BBL_USE=F`, and `SSW_CALC_ZNOT=F`.

The resulting analysis is archived as `v003_case_a_sediment_on_no_bbl`. It showed almost no change relative to the sediment-on case-a run with the original BBL settings: case a ended at 573.28 kg in v003, compared with 582.41 kg in v002. This is important negative evidence: the sediment/plastic interference is unlikely to be explained only by the Sherwood/Signell/Warner BBL roughness feedback. It also means that either the HPC build did not actively use that BBL path, or another sediment-enabled code path dominates the plastic difference.

New Isolation Test: Delayed Advance_Sed

A sharper test was then prepared using `generic_sediment_a_inactive.inp`. This keeps `SEDIMENT_MODEL=T` and the sediment setup path active, but sets `SED_START = 999999999`. In `mod_sed.F:883`, `Advance_Sed` returns immediately when `iint < sed_start`. Therefore this test keeps sediment initialization, the sediment/plastic compile path, and the river-forcing optional-argument path, but removes active runtime sediment transport.

The recent partial run is already diagnostically useful: the `total_mass_by_case.png` curve is reported to overlay the original v001 behavior. Assuming the completed run preserves this result, the main interference is not from sediment setup, sediment namelist presence, or the compile-time river-forcing path. It is instead localized to code executed after the early return in `Advance_Sed`, i.e. the active sediment operator.

Cap, Bed, and Split Checks

The hard concentration cap remains inactive in the present outputs. The maximum extracted concentrations are far below the code cap of 100 kg m^{-3} :

$$\max C_a = 2.41 \times 10^{-4}, \quad \max C_b = 6.98 \times 10^{-3}, \quad \max C_c = 1.45 \times 10^{-2} \text{ kg m}^{-3}.$$

The cap-threshold counters are zero for all cases with available counters. Thus `mod_plast.F:1242--1243` is still a non-conservative code site, but it is not active in this 10-day test.

For the null-flocc comparison, bed exchange is also not the leading cause. Cases a and b both end with zero sampled bed MP1 mass and zero sampled deposition/erosion exchange. Case c does exchange with the bed, so it should be interpreted separately.

The saved flocc split fields are internally consistent:

$$\text{mp1_agg} + \text{mp1_dis} \approx \text{mp1}.$$

The maximum total split-minus-combined mismatch is zero for case **b** and 1.83×10^{-7} kg for case **c**. Saved-output recombination is therefore not the present sink.

Sediment/Plastic Call Trace

The setup and timestep order make the sediment-runtime finding plausible.

1. **Setup order.** In `fvcom.F:643--651`, `SETUP_SED` is called before `SETUP_PLAST`. Thus the sediment module is initialized first when `SEDIMENT_MODEL=T`.
2. **Timestep order.** In `internal_step.F:1092--1137`, `ADVANCE_SED` is called before `ADVANCE_PLAST`. The earlier offline-style block in `internal_step.F:213--235` has the same ordering. Plastic therefore sees the post-sediment state each internal step.
3. **Sediment dynamics.** `mod_sed.F:859--1156` advances active-layer thickness, settling/deposition, erosion, bed stratigraphy/bottom properties, horizontal advection, FCT limiting, vertical diffusion, OBC/source boundary conditions, nonnegative/upper clipping, and sediment exchange.
4. **Plastic dynamics.** `mod_plast.F:914--1851` computes ambient density and flocc rate fields, updates plastic active-layer quantities, deposition/erosion/update-bottom, horizontal advection, vertical diffusion, kinetics, OBC/source boundary conditions, clipping, MPI exchange, and final recombination.
5. **Flocc coupling to sediment.** In `mod_plast.F:4619--4815`, the aggregation rate calculation reads `sed(ised)%conc`, `sed(ised)%Sd50`, and `sed(ised)%Wset`. Thus active flocc interaction is explicitly tied to sediment size and suspended sediment concentration. In case **b**, the saved output still has `mp1_agg=0`, so the current **b** run is effectively a split-disaggregated/null-flocc test despite executing the flocc branch.

Diagnosis

1. **Sediment runtime changed domain retention.** The drop from 1001.25 kg in `v001` to 582.41 kg in `v002` shows that turning on the sediment model changes the transport environment and/or open-boundary export enough to strongly affect domain MP1 mass. This is not necessarily a plastic mass sink by itself; the plotted domain mass does not include cumulative open-boundary export.
2. **BBL roughness is now demoted.** The archived `v003 generic_sediment_a_no_bbl.inp` test changed case-a final mass by only 9.13 kg relative to the `v002` BBL-on sediment case. Bottom-stress feedback should remain documented, but it is no longer the leading explanation for the present discrepancy.
3. **The no-flocc plastic path itself changes when sediment is compiled/enabled.** In `mod_plast.F`, the update enters a `#if defined(SEDIMENT)` block before checking `flocc_intract`. Even with `PLAST_FLOCC=F`, case **a** uses the sediment-build no-flocc branch

around `mod_plast.F:1248--1278`. Therefore an old case-a run built without `-DSEDIMENT` is not a strict code-path match to a sediment-build case-a run.

4. **River forcing/source updating is now demoted.** The river update call still changes at compile time: with plastic only, `bcond_gcn.F:645` calls `UPDATE_RIVERS` with `PLAST=PLASTDIS`; with sediment and plastic, `bcond_gcn.F:648` calls it with both `SED=SEDDIS` and `PLAST=PLASTDIS`. However, the `delayed-Advance_Sed` test keeps this compile/setup path while recovering the `v001`-like mass curve. Thus river optional-argument handling is unlikely to be the primary cause of the present case-a discrepancy, although cumulative river MP input should still be logged for closure.
5. **The leading suspect is now active `Advance_Sed` side effects.** `Advance_Sed` is called before `Advance_Plast` each internal step. With `MORPHO_MODEL=F`, bathymetry should not be updated, and with BBL disabled the roughness route is weakened. The `delayed-Advance_Sed` test localizes the main discrepancy to the active sediment operator, especially the shared scalar-transport calls and any shared model state they leave behind before plastic advances.
6. **Shared scalar-transport state is a concrete mechanism to test.** Inside `mod_sed.F:1016--1028`, sediment calls `adv_scal_backward` and `fct_sed`. The shared scalar routine writes global bottom-gradient arrays `PFPXB/PFPYB` in `mod_scal.F:911--912`. No-floc plastic should not directly read sediment concentration, so a case-a response to active sediment transport points toward shared-state contamination or another common scalar side effect rather than physical floc aggregation.
7. **A sediment exchange mismatch can affect floc cases.** After sediment clipping, `mod_sed.F:1099--1104` updates `sed(i)%conc`; the following MPI exchange at `mod_sed.F:1117--1122` exchanges `sed(i)%cnew`, not the clipped `sed(i)%conc`. Floc-rate calculations later read `sed(ised)%conc`. This is less likely to explain the no-floc case-a effect by itself, but it is a strong candidate for b/c rank-boundary oscillations and should be corrected or instrumented.
8. **The remaining case-a/case-b gap is still a split-path problem after sediment state is matched.** With `SEDIMENT_MODEL=T` in both cases, the remaining 210.56 kg deficit in case b points back to differences between combined `conc` and split `conc_d` operators.
9. **The leading code-level suspects are water-column transport and boundary handling.** With cap counts zero and no sampled bed storage in case b, the likely divergence occurs during split `adv_scal`, split `vdif_scal`, split `bcond_scal_OBC`, split `bcond_scal_PTsource`, or nonnegative clipping before output.
10. **The split branch still has one awkward exchange point.** Before OBC handling, the floc branch exchanges `cnew_a`, `cnew_d`, and combined `cnew` (`mod_plast.F:1513--1516`). At that point the meaningful prognostic state is split; combined `cnew` is not rebuilt until `mod_plast.F:1715--1716`. This may be harmless, but it is not strictly symmetric with the no-floc path.
11. **Final recombination improved the budget but did not solve it.** The patched final order now clips and exchanges `cnew_a/cnew_d`, then rebuilds `conc = cnew = conc_a+conc_d`. The mass recovery after that change confirms it addressed a real asymmetry, but the remaining deficit shows another divergence exists upstream.

12. **Direct floc-rate reads from sediment should be bypassed in a true null-kinetics test.** The aggregation/disaggregation routines read `sed(ised)%conc`, `Sd50`, and `Wset`. For a strict “kinetics off” symmetry test, $\lambda_a = \lambda_d = 0$ should be imposed without calling sediment-dependent rate calculations at all. That removes any possible sediment-size or sediment-concentration dependence from case b.
13. **Case c is not a pure conservation comparator.** Case c retains much more total mass than case b, but it has active split components and bed storage. It is useful for full floc behavior, while case b remains the sharper test for whether split `conc_d` can reproduce combined `conc`.

Focused Next Checks

1. Use `v003` as the latest pre-inactive-`Advance_Sed` archive for future plots and memo updates. If the delayed-`Advance_Sed` run finishes with the same `v001`-like behavior, archive that analysis as `v004` and record it as the decisive “sediment initialized but runtime operator inactive” comparison.
2. Bisect `Advance_Sed` using temporary early returns. First return after `calc_active_layer`; then after `calc_deposition/calc_erosion/bed` updates but before `adv_scal_backward`; then after `adv_scal_backward/fct_sed`; then after `vdif_scal/MPI` exchange but before `OBC/source/clipping`. This should pinpoint which block first changes plastic-domain mass.
3. Add a targeted shared-state checksum immediately before and after `Advance_Sed`, before `Advance_Plust`. Track small summaries for `PFPXB`, `PFPYB`, `D`, `DT`, `DZ`, `KH`, `U`, `V`, `UARD_OBCN`, and the shear-stress array passed to plastic. Any non-sediment state changed by `Advance_Sed` becomes an immediate smoking gun.
4. Run a narrow save/restore test around sediment `adv_scal_backward` for `PFPXB/PFPYB`. If restoring these arrays removes the case-a sediment-runtime effect, the coupling mechanism is shared scalar gradient state.
5. Test `SED_ONED=T` as an intermediate isolation. This skips the horizontal sediment advection/`OBC/source` path in `Advance_Sed` while leaving more of the vertical/bed sediment machinery active. A return toward `v001` would implicate the horizontal scalar transport block.
6. Add a river-source budget log for `MP1`. At each report interval, record instantaneous $\sum_r Q_r C_{\text{plast},r}$ and cumulative $\sum_t \sum_r Q_r C_{\text{plast},r} \Delta t$. This will determine whether the apparent domain-mass loss is actually lower source input or stronger export.
7. Add a short internal checkpoint diagnostic for one small matched run. Record domain-integrated mass for `conc`, `conc_d`, and `conc_a+conc_d` after: start, deposition, erosion/update bottom, `adv_scal`, `vdif_scal`, `OBC`, point-source BC, nonnegative clamp, `MPI` exchange, and recombination.
8. For the null-floc case, add a temporary switch or guard so $\lambda_a = \lambda_d = 0$ is assigned directly and the sediment-dependent floc rate routines are not called.
9. For b/c, reconcile sediment post-clipping exchange: exchange `sed(i)%conc` after clipping, or set `sed(i)%cnew = sed(i)%conc` before exchanging. This should be treated as a diagnostic fix first, because it affects the sediment state read by the floc-rate routines.

10. Explicitly accumulate mass removed by nonnegative clipping. The hard cap should remain zero; any nonzero clipping loss would be immediately informative.
11. If serial and parallel runs differ, focus on the two exchange locations: `mod_plast.F:1513--1516` before OBC and `mod_plast.F:1677--1716` at final recombination.